Trishir Kulkarni

parametric CAD / design for manufacturing / drawings & GD&T

Aalen, Germany — relocating to Munich
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Portfolio: [insert link]

AVAILABILITY Enrolled M.Sc. mechanical engineering student · available 20 h/week during semester and full-time in semester breaks · relocating to Munich, on-site · English C2, German A2

PROFILE

Mechanical engineer whose working year was spent producing CAD that other people had to build from. I model parametrically because the parts I work on get revised — a feature tree that survives a dimension change is the difference between an edit and a rebuild. My hardest modelling job was a cycloidal gear profile driven directly from its design equations, where eccentricity and roller pitch are linked parameters and changing one has to propagate correctly through the rest. Alongside that, my research work has been almost entirely about the same question this role asks of an AI model: does the thing we computed actually hold up when checked against reality?

PARAMETRIC CAD & MANUFACTURING

09/2023 – 08/2024

Mechanical Design / FEA Engineer — Zeus Numerix Pvt. Ltd., Pune

Design and simulation services for defence and aerospace clients

- Released production-ready parts by building parametric SolidWorks models and complete drawing packages — dimensioned and GD&T-toleranced drawings, assembly documentation and bills of materials — under PDM revision control, structured so colleagues could modify and re-release them without rebuilding.
- Reduced post-tooling iterations by applying design-for-manufacturing constraints, fits and tolerance stack-up at concept stage, and checking the intent held when machinists raised issues on the shop floor.
- Confirmed parts met their structural and thermal margins by running FEA in ANSYS and comparing predictions against measured data before release.

08/2022 – 01/2023

Mechanical Design Intern (B.E. dissertation) — DRDO R&DE (Engineers), Pune

Two-stage cycloidal speed reducer · three-person team

- Modelled an equation-driven cycloidal disc profile in SolidWorks, where rolling-circle diameter, eccentricity and roller pitch form one linked parameter set — changing any one to suit packaging forces the others, so the model had to be built to propagate correctly.
- Carried tolerance stack-up and manufacturability analysis across every mating precision component before drawing release, with eccentricity as the dimension the assembly depends on.
- Wrote MATLAB routines computing contact-force distribution across the housing rollers as a function of input angle, as a calculation tool for the team.

02/2021 – 03/2022

Suspension Engineer — Stallion Motorsports Electric (Formula Student), Pune

Student motorsport team, electric vehicle category

- Designed the complete suspension in SolidWorks, produced the drawings, then machined, laser-cut and 3D-printed the parts in-house — so every modelling decision came back to me as a real part on a bench.
- Chose the manufacturing process per component (CNC, laser cutting, 3D printing, off-the-shelf) based on what each could actually deliver in accuracy, stiffness and material.
- Built locating and drilling jigs for repeatable assembly, and checked fitment dimensionally before anything was bonded or welded.

MODEL EVALUATION & VALIDATION

since 2025

Master's thesis — Institute of Drive Technology, Hochschule Aalen

BMBF research project · supervised by Prof. Kley

- Build and test numerical lifetime models for drivetrain components against ISO 281, DIN 743 and DIN 3990, implemented in MATLAB, and establish where each model is and is not trustworthy.
- Validate computed results against independent hand calculations and run sensitivity analyses on the governing parameters; report findings in structured written form to a supervisor.
- Research paper on bearing life arising from this work currently under review.

2022 – 2023

Instrumented test bench — measurement against simulation

Five-person team · published in JETIR, Vol. 10 Issue 5, 2023

- Built and instrumented a measurement rig, then compared measured against predicted results across a full test matrix to establish whether the simulation deserved to be believed — the two agreed within small error.

EDUCATION

10/2024 – 09/2027	M.Sc. Advanced Materials and Manufacturing — Hochschule Aalen, Germany Manufacturing process theory: machining, casting, injection moulding, stamping
08/2020 – 07/2023	B.E. Mechanical Engineering (Hons. Electric Vehicles) — SPPU, Pune Final grade 1.9 (German scale)
06/2017 – 06/2020	Diploma in Mechanical Engineering — MGM's Polytechnic, Aurangabad Final grade 1.78 (German scale)

TOOLBOX

cad	SolidWorks — parametric part and assembly modelling, configurations, drawings, BOMs, PDM revision control · CATIA V5 (basic) · AutoCAD · comfortable picking up Onshape, Fusion or NX
manufacturing	Design for manufacturing · CNC machining · laser cutting · 3D printing and slicer software · drill press · jigs and fixtures · assembly and fitment checking
drawings	GD&T · tolerance stack-up · fits and clearances · assembly documentation · specification lists · drawing revision control
scripting	Python (basic scripting and data evaluation) · MATLAB (calculation tools, parameter sweeps) · MS Excel
analysis	ANSYS — structural, thermal, modal · FEM · measured-vs-predicted correlation · hand-calculation cross-checks
publications	JETIR 10(5), 2023 · IRJET 07(06), 2020 · bearing-life paper under review